Q1

a) $\lim\limits_{n\to\infty} \dfrac{(n^2-3n)^2}{5n^3+n} = \dfrac{n^2(n-3)^2}{n(5n^2+1)} = \dfrac{n.(n-3)^2}{5n^2+1} = \dfrac{n^3-6n^2+9n}{5n+1}$

l'hospital

$\lim\limits_{n\to\infty} \dfrac{3n^2-12n+9}{5} = \infty \quad \therefore f(n) \in \Omega(g(n))$

b) $\lim\limits_{n\to\infty} \dfrac{n^3}{\log_2 n^4} \xrightarrow{\text{l'hospital}} \dfrac{3n^2}{\frac{1}{n^4 \ln 2}} = 3n^6 . \ln(2) = \infty$

$\Rightarrow \therefore f(n) \in \Omega(g(n))$

c) $\lim\limits_{n\to\infty} \dfrac{5n \log_2 4n}{n . \log_2 5^n} = 5. \lim\limits_{n\to\infty} \dfrac{n \log_2 4n}{n \log_2 5^n} = 5. \lim\limits_{n\to\infty} \dfrac{n . \log_2 4n}{n^2 . \log_2 5} = \dfrac{\log_2 4n}{n . \log_2 5}$

$5. \lim\limits_{n\to\infty} \dfrac{\log_5 4n}{n} \xrightarrow{\text{l'hospital}} 5. \lim\limits_{n\to\infty} \dfrac{1}{4n \ln(5)}$

$\dfrac{5}{4\ln(5)} \lim\limits_{n\to\infty} \dfrac{1}{n} = 0 \quad \therefore f(n) \in O(g(n))$

d) $\lim\limits_{n\to\infty} \dfrac{n^n}{10^n} = \lim\limits_{n\to\infty} \left(\dfrac{n}{10}\right)^n = \lim\limits_{n\to\infty} e^{n \ln\left(\frac{n}{10}\right)} = \infty \quad \therefore f(n) \in \Omega(g(n))$

e) $\lim\limits_{n\to\infty} \dfrac{8n \sqrt[5]{2n}}{n . \sqrt[3]{n}} = \lim\limits_{n\to\infty} \dfrac{8 \sqrt[5]{2} . \sqrt[5]{n}}{\sqrt[3]{n}} = \lim\limits_{n\to\infty} \dfrac{8 . \sqrt[5]{2} . \sqrt[15]{n^3}}{\sqrt[15]{n^5}}$

$\lim\limits_{n\to\infty} \dfrac{8\sqrt[5]{2}}{n^{2/15}} = 0 \quad \therefore f(n) \in O(g(n))$

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Q2

a) static void methodA (String str_array[]){
for (int i = 0; i< str_array.length ; i++) → n
str_array[i] = ""; → 1
}

→ constant will be ignored.

$O(n \times 1) = O(n)$ //

b) Static void methodB (String str_array[]) {
for (int i = 0; i< str_array.length; i++) → n
methodA(str_array); → n
for (int j=0; j< str_array.length; j++) → n
System.out.println (str_array[j]); → 1
}

→ recessive one will be ignored.

$O(n \times n) + O(n) = O(n^2)$

c) Static void methodC (String str_array[]) {
for(int i=0; i<str_array.length; i++) → n
for(int j = 0; i< stry-array.length; j++) → n
methodB (str_array); → n^2
}

$O(n \times n \times n^2) = O(n^4)$ //

d) static void methodD (String str_array[]){
for(int i = 0; i< str_array.length; i++){ → n
System.out.println(str_array[i]); → 1
str_array [i--] = ""; → 1
}
}

inside of the firstly i is increasing then in the fourth line of the code it's decreasing. That's an infinite loop.

It can't be calculated.

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e) static void methodE(String str_array[]) {

```
    for(int i=0; i< str_array.length ; i++) → n
        if(str_array[i] == " ") → 1
            break;
}
```

$O(n \times 1) = O(n)$ //

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Q3

a) Sorted Array :

```
function sortedMax(A):
    if array A's length less than 2
        return -1
    else
        return A[A.length-1] - A[0]
```

it means that if we calculate the time complexity there are two conditions if array length less than 2 it means that there isn't any max value to be calculated.
else if there are more than 2 index in the array it means according that array is sorted first element is the min value the last eleman is the max value. The max difference is
A[n-1] - A[0]
The statements have 1 time complexity

$O(1+1) = O(2) = O(2 \times 1) = O(1)$
↳ we ignore the constant.

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b) Unsorted Array:

function unsortedMax(A):

max assigns A[A.length-1]
min assigns A[0]

for int i from 0 to A.length ; i increases 1 :

if max is less than A[i] :

set max assigns A[i]

if min is greater than A[i] :

set min assigns A[i]

return max minus min

if the length of Array A is less than 2 :

return -1

Firstly I assign the min and max values to a random index then I do a loop I checked with the if statements if there is a number is greater than max, sets this number to a max.
if there is a number is less than min, sets this number to a min.
finally return the differences of max and min
else if the numbers in the array less than 2, executes the code.

Time complexity;

$O(n \times (1+1)) + O(1) = O(n \times 2) + O(1) = O(n)$ //

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